Camila Paz Guajardo Vásquez

cami-gv | Camila Guajardo Vásquez | cguajardo@dim.uchile.cl

PhD student
Department of Mathematical Engineering, Universidad de Chile.

Research interests in geometric topology, geometric analysis and hyperbolic geometry. She has experience on design and implementation of algorithms, project management and scientific communication for varied audiences. Academic career recognized with Chilean's government scholarships.

Education

- PhD in Engineering Sciences, Mathematical Modeling — 2026 – present
  - Affiliation: Faculty of Physical and Mathematical Sciences, Universidad de Chile
  - Supervisors: PhD. Rayssa Caju and PhD. Pedro Montero Silva
  - Grants: ANID National Doctorate Schorlarship 2026
- Master of Science in Mathematics — 2024 – 2026
  - Affiliation: Faculty of Sciences, Universidad de Chile
  - Thesis: *On the action of the automorphism group of a compact Riemann surface on its set of systoles*
  - Supervisor: PhD. Anita Rojas Rodríguez
  - Grants: ANID National Master's Schorlarship 2025
- Bachelor's Degree in Mathematics — 2020 – 2023
  - Affiliation: Faculty of Mathematics, Pontificia Universidad Católica de Chile
  - Undergraduate thesis: *Mapping class groups of surfaces*
  - Supervisor: PhD. Mauricio Bustamante Londoño
  - Award: Academic Excellence, Class of 2023 (Bachelor's in Mathematics)

Research Experience

- Thesis Student, Fondecyt Project No. 12340034 — 2024
  - Topic: Group actions on compact Riemann surfaces and their effects on closed geodesics.
  - Results: Implementation of computational routines in Python/SageMath for building and analyzing complex geometric structures.
  - Supervisor: Prof. Anita Rojas Rodríguez
  - Repository: `https://github.com/Cami-GV/Hyperbolic-polygons`
- Undergraduate Research Projects (Vicerrectoría de Investigación, PUC) — 2022–2023
  - Riemann-Hurwitz formula and applications, with Prof. Natalia García Fritz
  - More theory and applications in topology, with Prof. Mauricio Bustamante Londoño
  - Geometry of plane curves, with Prof. Giancarlo Urzúa

Academic Activities (Selected)

- Geometric Analysis by the Andes (International conference). Universidad de Chile, Chile — 2026
  Poster presentation. Topic: An Algorithm to Provide Systolic Bounds from Group Actions
- XXV Coloquio LatinoAmericano de Álgebra. Pontificia Universidad Católica del Perú, Lima, Perú — 2026
  Short communication. Topic: Systolic Bounds from Hyperbolic Polygons.
- XXII Escola de Geometria Diferencial. Teresina, Piauí, Brazil — 2026
  Poster presentation. Topic: An algorithm to provide systolic bounds from hyperbolic polygons.
- XXXVII Jornadas de Matemática de la Zona Sur, Universidad del Bío-Bío sede Concepción, Chile — 2026
  Short communication. Topic: Systolic bounds from Hyperbolic Polygons.

- Organizer, Student Lectures Series in Riemannian Geometry, Pontificia Universidad Católica de Chile, Chile 2025
  Topics: Smooth manifolds, tangent spaces, Riemannian metrics, model spaces, connections, geodesics, and curvature.
  Faculty Advisor: Prof. Pedro Gaspar

- Student's Seminar, Universidad de Chile, Chile 2024–2025
  Talks included *Concrete surfaces using SageMath* and An Introduction to Morse theory

- Number Theory Seminar, Pontificia Universidad Católica de Chile 2022-2023
  Talks on additive fibers and elliptic surfaces over $\mathbb{P}^1$,and elliptic curves and their rational points

Selected Coursework

- POST664 – Analysis II (Prof. Nikola Kamburov) 6.4/7.0
  Contents: Banach and Hilbert spaces, Baire and Hahn–Banach theorems, Lax–Milgram theorem, spectral theory, Sobolev spaces, distributions

- POST5900 – Introduction to Riemann surfaces and algebraic curves (Prof. Anita Rojas) 7.0/7.0
  Contents: Riemann surfaces, algebraic curves in $\mathbb{P}^2(\mathbb{C})$, meromorphic functions, holomorphic maps, Riemann–Hurwitz formula

- CSDMATE157 – Complex tori (Prof. Robert Auffarth) 6.1/7.0
  Contents: Torus homomorphisms, Néron–Severi group, dual tori, extensions, subtori complements

- CS07MM0019 – Geometry and topology of manifolds (Prof. Mauricio Bustamante) 6.7/7.0
  Contents: Smooth manifolds, immersions and submersions, Sard's theorem, transversality

- MAT2555 – Functional analysis (Prof. Nikola Kamburov) 6.3/7.0
  Contents: Banach spaces, Baire's and Hahn–Banach's theorems, Hilbert spaces, Fourier series, spectral theory of self-adjoint compact operators

- MAT2850 – Algebraic topology (Prof. Siddarth Mathur) 6.3/7.0
  Contents: Covering spaces, fundamental groups, singular and cellular homology

- MAT2335 – Introduction to Algebraic geometry (Prof. Natalia García) 6.0/7.0
  Contents: Algebraic curves, affine and projective varieties, birational morphisms

- MAT2305 – Differential Geometry, con (Prof. Mauricio Bustamante) 6.4/7.0
  Contents: Curves in $\mathbb{R}^3$, fundamental forms, curvature, geodesics, Gauss–Bonnet's theorem

Teaching experience (Selected)

- Instructor. Department of Mathematics, Universidad Andrés Bello
  - Mathematical thinking for Nursing 2026
- Teaching Assistant. Faculty of Mathematics, Pontificia Universidad Católica de Chile
  - Geometry and topology of manifolds (graduate level), Prof. Pedro Gaspar 2025
  - Complex analysis, Prof. Martin Chuaqui 2024
  - Geometry I, Profs. María José Moreno and María Fernanda Espinal 2023
  - Abstract Algebra II, Prof. Ricardo Menares 2023
  - Introduction to Algebra, Prof. Amal Taarabt 2022
- Teaching Assistant. Department of Mathematics, Universidad de Chile
  - Algebra and Geometry II, Prof. Anita Rojas 2025
  - Algebra in Geometry I, Prof. Nelda Jaque Tamblay 2024

Skills

- Coding: Python, SageMath, LaTeX, and GitHub.
- Transversal competencies: technical communication, research project's management, mentoring.
- Languages: Spanish (mother tongue), and English (conversational).

Outreach and service

- Coach, Chilean team for International Mathematical Olympiads — 2021 – present
- Problem grader, Chilean National Mathematical Olympiad — 2021 – present
- Invited speaker, Inspiring talks in NiñasPro — 2026
  NiñasPro is a Chilean non-profit organization focused on empowering girls to pursue scientific careers.
- Mathematics Festival Monitor, Puerto de Ideas Antofagasta — 2023
  Sciences' outreach event for non-academic audiences.
- Member, Academic Committee CMat (Campeonato de Matemáticas) — 2020 – 2022
  CMat is a mathematics championship for secondary school students.
- Problem Coordinator, 1st Pan–American Girls' Mathematical Olympiad — 2021

Mathematical Olympiads' Participation

- Ibero-American MO, Bronze Medal — Mexico, 2019
- International MO, Honorable Mention — UK, 2019
- European Girls' MO, Bronze Medal — Ukraine, 2019
- International MO — Romania, 2018
- South Cone MO, Honorable Mention — Ecuador, 2017
- International MO — Brazil, 2017